Environmental studies

Unit 2: Natural Resources

Topics to be covered: Renewable and non-renewable resources: Natural resources and associated problems. Forest resources: Availability, Depletion of Forests, Environment impact of forest depletion on society. Water resources: Use and over-utilization of surface and ground water, floods, drought, etc. Mineral Resources: Use and exploitation, environmental effects of extracting and using mineral resources. Energy Resources: Growing energy needs, renewable and non-renewable energy sources, and use of alternate energy sources. Land Resources: Land as a resource, land degradation, man induces landslides, soil erosion, and desertification.

Renewable and non-renewable resources:

Ecosystems act as resource producers and processors. Solar energy is the main driving force of ecological systems, providing energy for the growth of plants in forests, grasslands and aquatic ecosystems.

The word resource means a source of supply. The natural resources include water; air, soil, minerals, coal, forests, crops and wildlife are examples. All the resources are classified based on quantity, quality, re-usability, men's activity and availability.

Natural resources are naturally occurring substances that are considered valuable in their relatively unmodified (natural) form. A natural resource's value rests in the amount of the material available and the demand for it. The term was introduced to a broad audience by E.F. Schumacher in his 1970s book Small is Beautiful.

a) Renewable resource or inexhaustible resources

The renewable resources can maintain themselves or can be replaced if managed wisely. These resources are constantly renewed in nature. The renewable resources are therefore not likely to be lost due to excessive and unwise use.

b) Non-renewable resources or exhaustible resources

These resources once used are lost forever, as they are not restored. They include metallic minerals and fossil fuels. At current rates of usage, all the industrial metals may lose for less than a century and those of petroleum and natural gas may exhaust in 15-20 years.

Natural resources and associated problems

Human population is growing day-by-day. Continuous increase in population caused an increasing demand for natural resources. Due to urban expansion, electricity need and industrialization, man started utilizing natural resources at a much larger scale. Non-renewable resources are limited. They cannot be replaced easily. After some time, these resources may come to an end. It is a matter of much concern and ensures a balance between population growth and utilization of resources. This overutilization creates many problems. In some regions there are problems of water logging due to over irrigation. In some areas, there is no sufficient water for industry and agriculture. Thus, there is need for conservation of natural resources.

There are many problems associated with natural resources:

Forest resources and associated problems

- Use and over-exploitation.
- Deforestation.
- Timber extraction.
- Mining and its effects on forest.
- Dams and their effects on forests and tribal people.

Water resources and associated problems

- Use and overutilization of water.
- Floods, droughts etc.
- Conflicts over water.
- Dams and problems.

Mineral resource and associated problems

- Use and exploitation.
- Environmental effects of extracting and using minerals.

Food resources and associated problems

- World food problems.
- Changes caused by agriculture and over grazing.
- Effects of modern agriculture.
- Fertilizer-pesticide problems.
- Water logging and salinity.

Energy resources and associated problems

- Growing energy needs.

Land resources and associated problems

- Land degradation.
- Man-induced landslides.
- Soil erosion and desertification.

FOREST RESOURCES

Forests are one of the most important natural resources and a part of biosphere since these are natural assets on this earth. Forests predominantly composed of trees, shrubs, woody vegetation etc… Approximately 1/3rd of the earth's total land area is covered by forests. Forests are important ecologically and economically. Ecologically forests are to be considered as earth's lungs because they consume CO2 and release O2which is required for sustaining the life on this earth. The poisonous gas CO2 is absorbed by the trees of forests and reduces the global warming and helps to continue hydrological cycle, reduce soil erosion. Forest ecosystems are extremely good & hold a good quantity of water.

Economically forests provide timber, fodder to grazing animals, firewood (conventional fuel), bamboos, rubbers, medicines, gums, resins, food items etc.

USES OF FOREST

1. Watershed protection:

- Reduce the rate of surface run-off of water.
- Prevent flash floods and soil erosion.
- Produces prolonged gradual run-off and thus prevent effects of drought.

2. Atmospheric regulation:

- Absorption of solar heat during evapo-transpiration.
- Maintaining carbon dioxide levels for plant growth.
- Maintaining the local climatic conditions.

3. Erosion control:

- Holding soil (by preventing rain from directly washing soil away).

4. Land bank:

- Maintenance of soil nutrients and structure.

5. Local use - Consumption of forest produce by local people who collect it for subsistence – (Consumptive use)

- Food - gathering plants, fishing, hunting from the forest. (In the past when wildlife was Plentiful, people could hunt and kill animals for food. Now those populations of most Wildlife species have diminished; continued hunting would lead to extinction.)
- Fodder - for cattle.
- Fuel wood and charcoal for cooking, heating.
- Poles - building homes especially in rural and wilderness areas.
- Timber – household articles and construction.
- Fiber - weaving of baskets, ropes, nets, string, etc.
- Sericulture – for silk.
- Apiculture - bees for honey, forest bees also pollinate crops.
- Medicinal plants - traditionally used medicines, investigating them as potential Source for new modern drugs.

6. Market use - (Productive use)

- Most of the above products used for consumptive purposes are also sold as source of income for supporting the livelihoods of forest dwelling people.
- Minor forest produce - (non-wood products): Fuel wood, fruit, gum, fiber, etc.which are collected and sold in local markets as a source of income for forest dwellers.
- Major timber extraction - construction, industrial uses, paper pulp, etc. Timber extraction is done in India by the Forest Department, but illegal logging continues in many of the forests of
- India and the world.

OVER EXPLOITATION OF FORESTS

Forest has been known to possess huge potential for human use and they have been exploited since early times for their vast potential. Exploitation of forest has taken place to meet human demands in the following ways:

- Due to wood cutting and large scale logging for raw materials like timber, pulp wood, fuel wood etc.
- Deforestation due to road construction
- Clearing of forest to create more agricultural lands to meet the food needs of growing population
- Encroachment of forests leading to destruction of about 19.57 lakh hectares (2013)of forest in the country
- About 78% of forest area is under heavy grazing
- Mining activities leads to clearing of forests
- Big hydroelectric projects result in large scale destruction of forest

In India, Joint forest management has come up as innovative approach involving community participation so that the rural economy is strengthened as well as forest resources are conserved through public involvement.

DEFORESTATION

Deforestation is the permanent destruction of indigenous forests and woodlands. The term does not include the removal of industrial forests such as plantations of gums or pines. Deforestation has resulted in the reduction of indigenous forests to four-fifths of their pre-agricultural area. Indigenous forests now cover 21% of the earth's land surface. Deforestation refers to the loss of forest cover (or) the aimless destruction of trees. The clearing of forests across the earth has been occurring on a large scale basis for many centuries. This process involves the cutting down, burning and damaging of forests. Currently 12 million hectares of forests are cleared annually and the current rate of deforestation continues, the world's forests will vanish within the next 100 years about 80% of the original forests on the earth have already been cleared.

Major causes of Deforestation:

- Shifting cultivation : There are an estimated 300 million people living as shifting cultivators who practice slash and burn agriculture and are supported so clear more than 5 lakh ha of forests for shifting cultivation annually. In India, we have this practice of North-East and to some extent in Andhra Pradesh, Bihar and M.P. which contribute to nearly half of the forest clearing annually.
- Fuel requirements: Increasing demands for fuel wood by the growing population in India alone has shooted up to 300-500 million tons in 2001 as compared to just 65 million tons during independence, thereby increasing the pressure on forests.
- Raw materials for industrial use: Wood for making boxes, furniture, railway-sleepers, plywood, match boxes, pulp for paper industry etc. have exerted tremendous pressure on forests. Plywood is in great demand for packing tea for Tea industry of Assam while fir tree wood is exploited greatly for packing apples in J & K.

- Development projects: Massive destruction of forests occurs for various development projects like hydroelectric projects, big dams, road construction, mining etc.
- Growing food needs: In developing countries this is the main reason for deforestation. To meet the demands of rapidly growing population, agricultural lands and settlements are created permanently by clearing forests.
- Overgrazing: The poor in the tropics mainly rely on wood as a source of fuel leading to loss of tree cover and the cleared lands are turned into the grazing lands. Overgrazing by the cattle leads to further degradation of these lands.
- Conversion of forests and woodlands to agricultural land to feed growing numbers of people.

Major activities and threats to Forests resources:

1.Timber Extraction: Logging for valuable timber, such as teak and Mahogany not only involves a few large trees per hectare but about a dozen more trees since they are strongly interlocked with each other a by vines etc. Also road construction for making approach to the trees causes further damage to the forests. The steps in timber extraction are:

a) Clear felling
b) Mechanized logging
c) Manual logging
d) Selective logging

2. Mining: Mining operations for extracting minerals and fossil fuels like coal often involves vast forest areas. Mining from shallow deposits is done by surface mining while that from deep deposits is done by sub-surface mining. More than 80000 ha of land of the country is presently under the stress of mining activities. Mining and its associated activities require removal of vegetation along with underlying soil mantle and overlying rock masses. This result in defacing the topography and destruction of the landscape in the area. Large scale deforestation has been reported in Mussorie and Dehradun valley due to indiscriminating mining of various minerals over a length of about 40 Km.

DAMS AND OTHER EFFECTS ON FOREST AND TRIBAL PEOPLE

Hydro-electric dams are main cause for deforestation. About 40,000 large dams are currently obstructing Workloads Rivers. Destruction of forest occurs for constructing big dams, which alters ecological balance. In these way landslides, droughts and floods conditions may rise in area. Socio-economic problems related to tribal and native people results from big dam construction Dam construction produces a number of health hazards. Thousands of workers who build the dams attacked by the diseases like measles, tuberculosis, syphilis etc. Dam building has resulted in wide range human rights violations. Rehabilitation policy of the government is important and typical when most of the displaced persons are tribal people. Tribal life and culture are mostly associated with forest.

WATER RESOURCES

Water resources are sources of water that are useful or potentially useful. Uses of water include agricultural, industrial, household, recreational and environmental activities. Virtually all of these human uses require fresh water.

Distribution of water on earth:

• 97% of the water on the Earth is salt water. Only 3% is fresh water; slightly over two thirds of this is frozen in glaciers and polar ice. The remaining unfrozen fresh water is found mainly as groundwater, with only a small fraction present above ground or in the air Fresh water occurs mainly in two forms

1. Ground water and
2. Surface water

Groundwater: About 9.86% of the total fresh water resources are in the form of groundwater and it is about 35-50 times that of surface water supplied.

USES OF WATER:

- DOMESTIC USE: Water used in the houses for the purposes of drinking, bathing, washing Clothes, cooking, sanitary & other needs. The recommended value according to Indian standard specification for domestic use is 135 liters/day
- INDUSTRIAL USE: Water is required for various industries such as cement, mining, textile, leather industries.
- PUBLIC USE: This includes water used for public utility purpose such as watering parks, Flushing streets, jails etc.
- FIRE USE: Water is used in case of accidents and to prevent the fire issues.
- IRRIGATION: To grow crops which is the main sources for food?
- OTHER USES: Hydroelectric power generation requires water.

OVER UTILIZATION OF GROUND WATER AND SURFACE WATER

Over use of groundwater has following effects.

1. Lowering of water table: Excessive use of ground water for drinking, irrigation and Domestic purposes has resulted in rapid depletion of ground water in various regions leading to lowering of water table & drying of wells.

The reasons for shortage of water are:

- Increase in population,
- Increasing demand of water for various purposes.
- Unequal distribution of fresh water.
- Increasing pollution of water sources cause over exploitation.

2. Ground subsidence: When ground water withdrawal is greater than its recharge rate, the sediments in the aquifer become compacted. This is called ground subsidence which may cause damage of buildings, destroy water supply systems etc.

3. Drought. A drought is an extended period of months or years when a region notes a deficiency in its water supply whether surface or underground water. Generally, this occurs when a region receives consistently below average precipitation.

We can define drought in four main ways:

- Meteorological drought: related to rainfall amounts
- Hydrological drought: determined by water levels in reservoirs
- Agricultural drought: related to the availability of water for crops
- Socioeconomic Drought: related to demand and supply of economic goods

a) Meteorological Drought: Meteorological drought is generally defined by comparing the rainfall in a particular place and at a particular time with the average rainfall for that Place. The definition is, therefore, specific to a particular location. Meteorological drought leads to a depletion of soil moisture and this almost always has an impact on crop production.

b) Hydrological Drought: Hydrological drought is associated with the effect of low rainfall on water levels in rivers, reservoirs, lakes and aquifers. Hydrological droughts usually are noticed sometime after meteorological droughts. First precipitation decreases and, Sometime after that, water levels in rivers and lakes drop.

c) Agricultural Drought: Agricultural drought mainly effects food production and farming. Agricultural drought and precipitation shortages bring soil water deficits, reduced ground water or reservoir levels, and so on. Deficient topsoil moisture at planting may stop germination, leading to low plant populations.

d) Socioeconomic Drought: Socioeconomic drought occurs when the demand for an economic good exceeds supply as a result of a weather-related shortfall in water supply. The supply of many economic goods, such as water, forage, food grains, fish, and hydroelectric power, depends on weather. Due to variability of climate, water supply is sufficient in some years but not satisfactory to meet human and environmental needs in other year.

FLOODS

A flood is an overflow of water that submerges land which is normally dry. The European Union (EU) Floods Directive defines a flood as a covering by water of land not normally covered by water. Flooding may occur as an overflow of water from water bodies, such as a river or lake, in which the water overtops or breaks, resulting in some of that water escaping its usual boundaries, or it may occur due to an accumulation of rainwater on saturated ground in an area flood. Floods can also occur in rivers when the flow rate exceeds the capacity of the river channel, particularly at bends in the waterway. Floods often cause damage to homes and businesses if they are in the natural flood plains of rivers.

MINERAL RESOURCES

A mineral is a naturally occurring substance of definite chemical composition and identifiable physical properties. An ore is a mineral or combination of minerals from which a useful substance, such as a metal, can be extracted and used to manufacture a useful product. The geological processes are caused for the formation of the minerals over millions of years ago in the earth's crust. Minerals are generally localized in occurrence and the deposits are very

sporadic in distribution. Mineral resources are non-renewable and the mineral /ore are extracted by the process of mining. Iron, aluminium, zinc, manganese and copper are important raw materials for industrial use. Important non-metal resources include coal, salt, clay, cement and silica. Stone used for building material, such as granite, marble, limestone, constitute another category of minerals. Minerals with special properties that humans value for their aesthetic and ornamental value are gems such as diamonds, emeralds and rubies. The lustre of gold, silver and platinum is used for ornaments. Minerals in the form of oil, gas and coal were formed when ancient plants and animals were converted into underground fossil fuels.

Uses of minerals:

Minerals are used in a large number of ways for domestic, industrial, commercial
Sectors etc…

1. Generation of energy by using coal (lignite / anthracite); uranium, gold, silver, platinum, diamond are used in jewellery. Copper, aluminium etc. are used as cables for transmission of power.
2. Some of the minerals are used in Ayurveda as medicine. Gold is reputed to strengthen the heart muscle and increase energy and stamina.

Mining and its Process:

Minerals and their ores need to be extracted from the earth's interior so that they can be used.

This process is known as mining. Mining is the extraction of valuable minerals or other geological materials from the earth, from an ore body, lode, vein, (coal) seam or reef, which forms the mineralized horizon and package of economic interest to the miner.

Mining operations generally progress through four stages:

- Prospecting: Searching for minerals.
- Exploration: Assessing the size, shape, location,
- Development: Work of preparing access to the deposit so that the minerals can be extracted from it.
- Exploitation: Extracting the minerals from the mines.

Environmental effects:

Mineral extraction and processing in mines involves a negative impact on environment. Much risk is involved in mining process because of high temperature, pressure Variations, fire hazards and lack of ventilation in mines.

• Mining process involves removal of over burden of soil, ore extraction & transportation, crushing & grinding of ore, water treatment of ore, and storage of waste material. As a result of these activities cause air pollution, noise pollution, water pollution, loss of habitat of wildlife, concentration of toxic substances in tailing ponds and spreading of dust.

• People working in mines often suffer from serious respiratory system and skin diseases.

• Mining often causes ground subsidence which results in tilting of buildings, cracks in houses, buckling of roads, bending of rail tracks etc.

• Exploration process before a mining involves, geochemical, geophysical surveys. Drilling activities which causes for air pollution, noise pollution etc...

• In addition, disturbance of all vegetation (flora) and fauna (animals) from that a region.
• Acid mine drainage (AMD), or acid rock drainage (ARD): The outflow of acidic water from (usually abandoned) metal mines or coal mines. However, other areas where the earth has been disturbed (e.g. construction sites, subdivisions, transportation corridors, etc.) may also contribute acid rock drainage to the environment.

ENERGY RESOURCES

- Energy is defined by physicists as the capacity to do work. Energy is found on our planet in a variety of forms, some of which are immediately useful to do work, while others require a process of transformation.
- Energy can neither be created nor destroyed but transformed from one form to other. Energy is closely related to force. When a force causes an object to move, energy is being transferred from the force to kinetic energy.
- Energy is present in a number of forms such as mechanical, thermal, chemical, biological energy etc..
- Energy production and utilization have become essential to carry out many activities in modern life.
- Energy is one of the important requirements that a country needs for its economic growth. At the same time, energy production has its impact on environment due to pollution and finally affects the quality of life of people.

GROWING ENERGY NEEDS

Energy plays a key role in the process of economic growth of a nation. The industrial development of any country is dependent on the organized development of its power resources'. Energy is also indispensable for agriculture, transport, business and domestic requirements. In fact, electricity has such a wide range of applications in modern economic development that its per capita consumption is, to a great extent, an index of the material advancement of the country. Energy is the capacity for doing useful work. It is an essential input for economic growth. This energy is used in the form of electrical energy, thermal energy, light, mechanical energy and chemical energy etc.

Energy is measured in joules in SI units. The annual per capita energy consumption in developed countries ranges from 5 to 11 kW whereas in the developing countries it is between 1 to 1.5 KW Only.

Uses of Energy

1. Energy is a primary input in any industrial operation.
2. It is also a major input in sectors such as commerce, transport, tele-communications etc.
3. The wide range of services required in the household and industrial sectors.
4. Owing to the far-reaching changes in the forms of energy and their respective roles in supporting human activities, research and training on various aspects of energy and environment have assumed great significance.

Types of energy: There are two main types of energy;

A. Non-renewable, and B. Renewable

A. NON – RENEWABLE ENERGY RESOURCES

Fossil fuels: Fossil means the remains of an animal or a plant which have become hard and turned into rock. All these found in earth's crusts which have been formed in the past by the geological processes. Fossil fuels are solid coal (lignite), liquid (crude oil / petroleum) and gases (natural gas).

a) Coal: Huge quantity of plant materials buried under earth's crust and altered by geological process and converted into carbon rich fuel. It is a non – renewable source because it takes a very long period (millions of years) for its formation. Coal is extracted by the process of mining and involves accidents due to mine collapse, ground water pollution, accumulation of poisonous material, explosive gases etc. cause diseases. CO2 pollution leads to greenhouse effect (global warming).
b) Crude oil: It is obtained in the form of liquid. The crude oil is heated up to 600oC in the oil refinery and condenses the vapours of hydro – carbons. Petrol petroleum products are refined fuels from crude oil. Petroleum products are used in large quantities in the manufacture of detergents, plastics, fertilizers, pharmaceuticals, synthetic rubber etc.. The transport sector consumes about 40% of diesel; 25% industries and 19%household and rest 16% agriculture and other sectors. .
c) Natural Gas: Gas deposits are trapped from the sedimentary formations by means drilling holes into the rock formations. While burning of natural gas, the emission of CO2 is less and thus reduces greenhouse effect and global warming. A total of 734 billion cubic mts of gas is estimated as proven reserves.

B. RENEWABLE ENERGY RESOURCES:

Renewable energy systems use resources that are constantly replaced and are usually less polluting. Examples include hydropower, solar, wind, and geothermal (energy from the heat inside the earth).

1. **Solar energy:** The energy which is derived from the sun is known as solar energy. It can be used for direct heating or sun's heat is converted into electricity. Photo voltaic cells convert direct solar energy into electricity.

A number of solar equipment have been developed to utilize sun rays to heat water, to cook food, to pump water and to run certain machines and used for street lighting, railway signals etc.

But the major problem with solar energy is that during cloudy weather it is available in less quantity than on sunny days.

How solar-panel works:

Solar panels are usually made from silicon, or another semiconductor material installed in a metal panel frame with a glass casing. When this material is exposed to photons of sunlight (very small packets of energy) it releases electrons and produces an electric charge. This PV charge creates an electric current (specifically, direct current or DC), which is captured by the wiring in solar panels. This DC electricity is then converted to alternating current (AC) by an inverter. AC is the type of electrical current used when you plug appliances into normal wall sockets.

Merits and demerits of solar energy:

- Solar energy is green energy i.e. it is renewable and does not cause pollution.
- Solar panels for the generation of electricity requires less maintenance.
- With the help of solar energy, electricity is possible to be generated even in the most remote, inaccessible locations where electric power lines cannot be laid.
- Electricity generation based on solar energy is weather dependent and hence it is less reliable. In winters and in cloudy days, the production becomes less.
- The whole set-up of the panels requires a lot of space to generate a considerable amount of electricity.
- The initial cost of a solar panel is considerably high.

2. **Hydro-Power energy:** Electrical power is generated by hydro-electric projects in which dams are constructed across the river. The kinetic energy of water is converted into mechanical energy by means of turbines and in turn, the mechanical energy is transferred into electrical energy by generators. Hydro power projects lead to several environmental problems like destruction of animal habitats, deforestation, migration of people etc..
3. **Geothermal energy:** Geothermal energy found within rock formations. Inside the earth the temperature rises with depth .The temperature in earth's crust is around 4000o C. Geysers (a natural spring that emits hot water) and hot springs are examples for geothermal energy where the steam and hot water come to the surface, in areas where the steam is tapped by drilling. The obtained steam is then used to generate power. Air pollution results in case of geothermal energy where the gases like H2S, NH3, CO2 present in the steam coming out of the geothermal sources.

The overall efficiency for power production is low (15%) as compared to fossil fuels (40%).

4. **Wind energy:** Wind energy is the kinetic energy associated with the movement of atmospheric air. Wind mills convert the wind energy into electrical energy. On an average wind mills can convert 30 – 40 % of available wind energy into electrical energy at a steady wind speed of 8.5mts / sec. The efficiency of wind mill is increased with the speed of wind and length of rotor blade.

The total wind energy potential in India's estimate is 25,000 MW of this about 6000 MW is located in Tamil Nadu; 5000 MW in Gujarat and contribute the states of Andhra Pradesh, Maharashtra, Uttar Pradesh and Rajasthan for balance quantity.

Merits & demerits of wind energy:

1. It is a non – polluting and environment friendly source of energy.
2. It is a renewable energy available at free of cost
3. Power generation is cheaper with nil recurring expenses.
4. Wind mills are suitable to erect at on shore, remote and rural areas where wind blows with required intensity.
5. Favourable in geographic locations which are away from cities.
6. Wind turbine design, manufacturing, installation is complex due to varying atmospheric conditions.

7. Wind power doesn't suitable for large scale generation.

5. **Ocean energy:** Seas and oceans are large water bodies. Seas absorb solar radiation and large amounts of solar energy are stored in the tides and waves of the ocean.

Ocean energy is non – polluting in nature and suitable at a few places only. Energy from seas or oceans is obtained from the following:

(1) Ocean Thermal Energy Conversion: The oceans collect and store huge quantities of solar on the surface of the water while the temperature of deep water is very low. Using this temperature difference it is possible to convert heat into electricity.

(2) Tidal energy: Tidal waves of the sea can be used to turn turbine and generate electricity.

Asia's first tidal power plant of 800 - 1000 MW capacity is proposed to be set up at Kandla in Gulf of Kutch.

Merits and Demerits of Tidal energy:

- Energy is freely available & inexhaustible.
- Power can be used to meet local needs of coastal area and for peak loads.
- Power is ensured round the year through its operation is intermittent.
- It is pollution free.
- It is unaffected by the vagaries of monsoon.
- It is site specific and the tidal energy can be recovered economically on the locations where the tidal range is 5m or more.
- The availability of tidal energy is variable thus the power generation is highly fluctuating in nature.
- Turbines are needed which can operate with fluctuating heads.
- The marine life and ecology is affected in the region where the tide plants are located. It also affects the navigation system.
- In order to handle large volume flow rates of tidal water at low heads, tidal plants need to operate with several turbines in parallel.
- Sea water is corrosive.

6. **Bio mass energy:** Bio-mass is an organic material from living beings or its residues. It is a renewable source of energy derived from the waste of various human and natural activities. The bio-mass energy sources include Wood, animal manure, sugarcane waste, agriculture crops, house hold waste, roots of plants, garbage etc. The simplest way of using bio-mass energy sources is to allow them to dry out in the sun and burn them.
7. **Bio-gas:** Bio-gas is a sustainable source of energy by virtue of its production from available natural organic wastes of cattle dung, human excreta, poultry waste, plant leaves, paddy husk etc…. Bio-gas is a mixture of methane (68%), CO_2 (31%) and N_2 (1%). Methane gas (CH_4) is produced by bio-gas plants and this gas is utilized as cooking gas whose calorific value varies from 4400 – 6200 Kilo Calories / cum. Heat value of biogas can be improved by reducing its CO_2 content. Bio-gas production is carried out in an enclosed bio-gas plant made of bricks or steel. Slurry of waste organic matter is fed into the plant through an inlet and gas formed is tapped by an inverted drum. As gas is produced the drum rises and the gas may be drawn through an outlet. Bio-gas is

commonly produced from cattle dung in a bio gas plant known as Gobar Gas plant. Bio-gas is a clean, cheap fuel that can be used for lighting purpose, lifting water through small pumps.

Merits and demerits of biogas:

- Non-polluting: Biogas burns without smoke; therefore, it evolves no harmful gas such as CO_2, CO, NO_2, and SO_2.
- Reduces Landfills: The slurry which produces after the production of biogas, we can use it as manure in fields. The technique of disposal is safe and effective and henceforth, no space gets wasted in the form of landfills.
- Inexpensive technology: A Biogas plant does not require an expensive installation cost and become self-sufficient within a time span of 3-4 months.
- Creates employment: It also creates work opportunity for thousands of people, especially in rural areas.
- Renewable source of energy: We consider it to be a renewable source of energy as the production depends on the production of waste which is an endless process.
- Inefficient on a large scale: As it is difficult to boost the efficiency of biogas, it is not economically feasible to use biogas on a large scale.
- Contains impurities: It comprises a lot of impurities which are difficult to control even after putting it through many rounds of purification. When we compress biogas to use as fuel, it proves to be extremely corrosive to the container.
- Unstable and hazardous: When methane comes in contact with oxygen it reacts violently to create carbon dioxide. As a result, the highly inflammable nature of methane makes it susceptible to explosions.

USE OF ALTERNATIVE ENERGY SOURCES

Alternative energy is any energy source that is an alternative to fossil fuel. These alternatives are intended to address concerns about such fossil fuels.

The nature of what constitutes an alternative energy source has changed considerably over time, as have controversies regarding energy use. Today, because of the variety of energy choices and differing goals of their advocates, defining some energy types as "alternative" is highly controversial.

In a general sense, alternative energy as it is currently conceived is that which is produced or recovered without the undesirable consequences inherent in fossil fuel use, particularly high carbon dioxide emissions, an important factor in global warming. Sometimes, this less comprehensive meaning of "alternative energy" excludes nuclear energy

• Solar energy is the generation of electricity from the sun. It is split up into two types, thermal and electric energy. These two subgroups mean that they heat up homes and generate electricity respectively.

• Wind energy is the generation of electricity from the wind.

• Geothermal energy is using hot water or steam from the Earth's interior for heating buildings or electricity generation.

• Biofuel and Ethanol are plant-derived substitutes of gasoline for powering vehicles.

• Nuclear binding energy uses nuclear fission to create energy.
• Hydrogen is used as clean fuel for spaceships, and some cars

Nuclear Energy or Atomic power: It is the energy which is trapped inside the atom. It is non–renewable source of energy which is released during fission or fusion of certain radioactive elements. The most important advantage of atomic power is the production of an enormous amount of energy from a small quantity of radioactive element. For eg: 1 kg of Uranium liberates energy equivalent to 30000 kg of coal.

Energy released during nuclear reaction (mass – energy equation as per Albert Einstein's formula E = mc2).

Nuclear Energy is produced by two processes namely (1) Nuclear Fission and (2) Nuclear Fusion.

Nuclear Fission: The nucleus in atoms is split by fast moving neutrons and in turn a tremendous amount of energy in the form of heat, light etc. is released by a chain of reactions. Uranium is used as fuel. The energy released slowly in this process is utilized to generate electricity or else released suddenly all at once, results a tremendous explosion as in the case of Atom bomb.

Nuclear Fusion: Nuclear energy can be generated by fusion process which involves two hydrogen atoms combine to produce one helium atom.

Eg: hydrogen bomb. The disposal of nuclear waste during mining, fuel production and reactor operation for a long time period results in adverse effects on environment. Disposable of nuclear waste is a national and global problem.

LAND RESOURCES

Land as a resource: Landforms such as hills, valleys, plains, river basins and wetlands include different resource generating areas that the people living in them depend on. Many traditional farming societies had ways of preserving areas from which they used resources. Eg. In the 'sacred groves' of the Western Ghats, requests to the spirit of the Grove for permission to cut a tree, or extract a resource, were accompanied by simple rituals. The outcome of a chance fall on one side or the other of a stone balanced on a rock gave or withheld permission. The request could not be repeated for a specified period.

Land on earth is as finite as any of our other natural resources. While mankind has learnt to adapt his lifestyle to various ecosystems world over, he cannot live comfortably for instance on polar ice caps, on under the sea, or in space in the foreseeable future.

Man needs land for building homes, cultivating food, maintaining pastures for domestic animals, developing industries to provide goods, and supporting the industry by creating towns and cities. Equally importantly, man needs to protect wilderness area in forests, grasslands, wetlands, mountains, coasts, etc. to protect our vitally valuable biodiversity.

Thus a rational use of land needs careful planning. One can develop most of these different types of land uses almost anywhere, but Protected Areas (National Park's and Wildlife Sanctuaries) can only be situated where some of the natural ecosystems are still undisturbed. These Protected Areas are important aspects of good land use planning.

LAND DEGRADATION

Soil is no less than our mother as it is indispensable for our survival. It is formed over long periods of time. But man is degrading it with his misdeeds. Many of our once-fertile soils have already been converted to agriculturally unfit alkaline or saline land or marshlands. It is estimated that there is more than 25 million hectares of such barren lands throughout the world. Our soil constitutes a biogeochemical shell around land and shallow waters. It is a product of the interactions of living matter with rocks. It profoundly affects the growth of living organism (especially plants) however; in turn it is influenced by the activities of the latter.

Reasons for Land Degradation

Various factors have led to Land Degradation. Some of them are as under:

1. Rapid increase in industrialization, urbanization and other activities or civilized man has exercised a tremendous impact on the soils and on other components of the biosphere.
2. Unplanned destruction of forests and forest litter has brought about serious change both in land and water.
3. The washing off of fine soil particles from deforested areas has caused great soil erosion.
4. Soil erosion has resulted in a great increase in run-off, pollution turbidity and mineralization in rivers and extensive silting in water reservoirs.

MAN-INDUCED LANDSLIDE:

Landslides always exist on this planet and the term land slide is used to describe a wide variety of process that result a downward movement of rocks under gravitational forces. In other words, mass movement of rocks, debris and soil down a slope of land.

Landslides are primarily associated with steep slopes. Surface run-off and changes in drainage also cause for landslides. Landslides can also be initiated by rainfall; earthquakes; volcanic activity, changes in groundwater movement or any combination these factors.

Debris-flows can travel down a hillside of speeds up to 200 miles per hour (more commonly, 30 – 50 miles per hour) depending on the slope angle, water content, and type of earth and debris in the flow.

While landslides are a naturally occurring environmental hazard they have recently increased in frequency in certain areas due to human activities.

Building excavations collapses in mining (e.g.: coal mine) causes landslides.

Simply landslides can be explained in three ways:

(a) Inherent of rocks (weakness in the structure of a rock)

(b) Due to heavy seismic or volcanic activity and

(c) Due to various environmental conditions.

SOIL EROSION:

The top layer of the earth is called as soil. Soil erosion occurs due to deforestation, overgrazing, industrialization; desertification etc.

a. Deforestation: Mining, industrial, urban development etc. causes deforestation and leads to exposure of the land to wind and rains causing soil erosion. Cutting trees leads to deforestation which in turn loss of organic matter in the soils.

b. Overgrazing: When sufficient amount of grass is available for the organisms usually the entire land /area may be subjected to exhaust and the land is exposed without grass and ultimately the land expose to wind/rain causing soil erosion.
c. Industrialization: Different processes carried out by industries and mining operations cause soil pollution which leads to degradation of land.

DESERTIFICATION:

Desertification is the process which turns productive into non- productive desert as a result of poor land-management. Desertification occurs mainly in semi-arid areas (average annual rainfall less than 600 mm) bordering on deserts. In the Sahel, (the semi-arid area south of the Sahara Desert), for example, the desert moved 100 km southwards between 1950 and 1975.

CAUSES OF DESERTIFICATION

* Overgrazing is the major cause of desertification worldwide. Plants of semi-arid areas are adapted to being eaten by sparsely scattered, large, grazing mammals which move in response to the patchy rainfall common to these regions. Early human pastoralists living in semi-arid areas copied this natural system. They moved their small groups of domestic animals in response to food and water availability. Such regular stock movement prevented overgrazing of the fragile plant cover.
* Cultivation of marginal lands, i.e lands on which there is a high risk of crop failure and a very low economic return, for example, some parts of South Africa where maize is grown.
* Destruction of vegetation in arid regions, often for fuel wood.
* Poor grazing management after accidental burning of semi-arid vegetation.
* Incorrect irrigation practices in arid areas can cause salinization, (the build-up of salts in the soil) which can prevent plant growth.

EFFECTS OF DESERTIFICATION

Desertification reduces the ability of land to support life, affecting wild species, domestic animals, agricultural crops and people. The reduction in plant cover that accompanies desertification leads to accelerated soil erosion by wind and water. South Africa is losing approximately 300-400 million tonnes of topsoil every year. As vegetation cover and soil layer are reduced, rain drop impact and run-off increases.

Water is lost off the land instead of soaking into the soil to provide moisture for plants. Even long-lived plants that would normally survive droughts die. A reduction in plant covers also results in a reduction in the quantity of humus and plant nutrients in the soil, and plant production drops further. As protective plant cover disappears, floods become more frequent and more severe. Desertification is self-reinforcing, i.e. once the process has started, and conditions are set for continual deterioration.